

AUTOMATED NETWORK REQUEST MANAGEMENT

ServiceNow Custom Application

Complete Project Documentation

Platform: ServiceNow

Project: Automated Network Request Management

Scope: Service Catalog, Approvals and Flow Designer Automation

Document: Implementation, Catalog, Automation and Testing Report

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1. Project Overview

The Automated Network Request Management project is a ServiceNow solution that lets employees and network teams submit, route, approve, and fulfil network-related service requests through a structured catalog record and Service Catalog experience. The implementation combines a custom network request table, catalog-driven form design, conditional UI Policies, an approval relationship, and Flow Designer automation.

- Requesters submit network requests (connections, access, devices) through a guided catalog form.
- Requests are stored in a custom network request table alongside standard platform tables.
- Conditional UI Policies reveal additional fields based on selections (e.g. connection type, device type).
- An Approval Request relationship links approval records (sysapprovals) to each network request.
- Flow Designer automates record creation, approval routing, email notification, and status updates.
- List views give approvers and network teams centralized visibility into requests, tasks, and statuses.

2. Problem Statement and Objectives

Network teams need a simple, consistent method to receive and process network-related service requests. A structured ServiceNow solution reduces unorganized, manual requests and gives network teams a consistent record containing requester and request information.

Objectives

- Streamline and automate the end-to-end lifecycle of network-related service requests.
- Reduce manual effort and human error in request intake and routing.
- Accelerate request fulfilment times.
- Enforce standardized workflows and approval processes.
- Improve visibility and tracking through a centralized platform.
- Enhance end-user experience with faster, more reliable service delivery.
- Ensure compliance with IT and security policies.

3. Application and Catalog Item Creation

The functional scope covers catalog creation, form design, approval routing, Flow Designer automation, and email notification. A Network Request catalog item is configured within the Service Catalog, mapped to the custom network request table, with stakeholder mapping and an execution roadmap defined during the design phase.

4. Custom Table and Data Architecture

The Automated Network Request Management solution is built on a structured data model that combines standard platform tables with a custom network request table for clarity, scalability, and control. Network requests are captured through catalog-driven records and stored with well-defined fields such as request type, access level, business justification, and related network configuration items.

Forms are designed to dynamically display only relevant information using UI logic, while mandatory fields enforce data accuracy and compliance. List views provide operational visibility for approvers and network teams, enabling efficient tracking of requests, tasks, and statuses. This architecture ensures secure access, seamless automation, and reliable reporting across the entire request lifecycle.

5. Approval Request Relationship

Creation of Approval Request on the Network Database Table

Activity 1: Creation of Related List

- Navigate to System Definition > Relationships.
- Click New to create a new relationship.
- Name: Approval Request.
- Applies to Table: the custom network request (database) table.
- Queries from Table: the sysapprovals table.
- Active: set to True.

- Save the relationship.

This relationship surfaces approval records directly on the network request form as a related list, giving requesters and approvers a single place to track approval status.

6. Form Design and Field Configuration

The Network Request form is designed to collect the information required to process a request, separating system-generated information (such as request number) from user-entered or selected information (such as request type, access level, business justification, and related network configuration items).

7. Catalog Variables Configuration

Variables are configured on the Network Request catalog item to build the form fields presented to the requester.

- Variable types include Single Line Text, Multi Line Text, Reference, and Choice, selected per requirement.
- Order values (100, 200, 300, ...) control the display sequence of fields.
- Question: the label shown to the requester.
- Name: the internal variable name used for scripting.
- Tooltip: guidance shown when the cursor hovers over the field.
- Example text: sample input to guide the requester.
- Mandatory / Read-only: configured as required per field.
- Auto-populate: dependent variables use dot-walking to pull related values automatically.

8. Navigation Flow

Procedure:

- Log in to the ServiceNow instance.
- Copy the instance domain, e.g. <https://dev190678.service-now.com>.
- Append the /sp prefix to reach the Service Portal, e.g. <https://dev190678.service-now.com/sp>.
- Search for "Network Requests".
- Fill in the required details and click Submit.
- A new request is generated with a request number, and the requester receives a confirmation email.

9. UI Policy and Conditional Validation

Catalog UI Policy Configuration

UI Policies display additional fields when certain choices are made, using conditional logic and field dependencies configured within ServiceNow.

Scenario 1: If the user selects "types of Connection" = Existing, the Existing ID field is displayed. Scenario 2: If the user selects "types of devices" = Others, the "Please specify" field is displayed.

Procedure:

- Navigate to the Network Request catalog item.
- Open the Catalog UI Policy related list and click New.
- Set Applies to = Catalog item, and Catalog item = Network Request.
- Provide a short description and the triggering condition (e.g. types of Connection is Existing).
- Save, then configure the UI Policy Action for the target variable and set it Visible = True.
- Update the UI Policy and test it on the catalog form.

10. Automation — Flow Designer

Flow Designer logic implements: Get Catalog Variables, Create Record, Ask for Approval, Send Email, Update Record — triggered on catalog item submission and supporting dynamic automation.

Activity 1: Creation of Flow

- Navigate to the Flow Designer home page.
- Click New to create a new flow.
- Provide the flow name: Network Request.
- Provide a description of the flow.
- Click Build flow, then add the Get Catalog Variables, Create Record, Ask for Approval, Send Email, and Update Record actions in sequence.

11. Access Control

Role-based access control restricts who can view, create, and update network request records and approvals, ensuring that only authorized requesters, approvers, and network team members can act on a given request.

12. QA Testing

Functional QA testing covered catalog submission, conditional UI Policy behavior, approval routing, automation execution, and record visibility for approvers and network teams.

13. Data Integrity

Mandatory fields, reference fields, and dependent/auto-populated variables enforce data accuracy across the request lifecycle, ensuring that records entering the network request table are complete and consistent.

14. Functional Flow



Figure 1: End-to-end functional flow — Requester to notification.

Step	Component	Function
1	Service Catalog	Requester submits the Network Request catalog item.
2	Catalog Variables	Capture request type, access level, justification and related items.
3	UI Policy	Display conditional fields based on connection/device type.
4	Custom Table	Store the network request record.
5	Approval Request Relationship	Link the request to its sysapprovals approval record.
6	Flow Designer	Create record, request approval, send email, update record status.
7	List Views	Give approvers and network teams visibility into requests and status.
8	Notification	Requester receives email confirmation and status updates.

End-to-end architecture: Requester → Service Portal → Network Request catalog item → Custom Table → Approval Request relationship → Flow Designer automation → Status-based tracking.

15. Testing and Validation

ID	Test	Expected / Evidence
TC-01	Submit network request	Request record is created in the custom network request table.
TC-02	Mandatory field validation	Form blocks submission until required fields are completed.
TC-03	Connection type = Existing	Existing ID field becomes visible per UI Policy.
TC-04	Device type = Others	"Please specify" field becomes visible per UI Policy.
TC-05	Approval relationship	Approval Request related list shows the linked sysapprovals record.
TC-06	Flow Designer execution	Get Catalog Variables, Create Record, Ask for Approval, Send Email and Update Record execute successfully on submission.
TC-07	Notification	Requester receives an email confirming submission.
TC-08	List view visibility	Approvers and network teams see the request and its current status.

A recorded walkthrough of the implementation is available at:

https://drive.google.com/file/d/1gDrM3CmFP848MBk_zRy7qVAJv1ECrgzL/view?usp=sharing

16. Conclusion and Future Enhancements

The Automated Network Request Management project demonstrates an end-to-end ServiceNow solution covering catalog-based intake, structured data storage, conditional form logic, approval tracking, and Flow Designer automation.

Key Achievements

- Structured data architecture combining standard and custom network request tables.
- Network Request catalog item with ordered, well-configured variables.
- Conditional UI Policies for connection type and device type scenarios.
- Approval Request relationship linking requests to sysapprovals records.
- Flow Designer automation covering the full request lifecycle.
- Role-based access control and validated data integrity.

Future Enhancements

- SLA-based escalation and automated reassignment for overdue requests.
- Deeper CMDB integration for network configuration items.
- Reporting dashboards for request volume, approval turnaround and fulfilment time.
- Expanded, multi-level approval hierarchies.
- Additional request types and knowledge-article recommendations during submission.

End of Project Documentation
Automated Network Request Management — ServiceNow